Even though his objections were shot down, Einstein remained deeply uncomfortable with quantum mechanics. In the following years he kept Bohr and his colleagues on their toes, leveling one new challenge after another. His most potent and far-reaching attack focused on something known as the *uncertainty principle*, a direct consequence of quantum mechanics, enunciated in 1927 by Werner Heisenberg.

Heisenberg and Uncertainty

The uncertainty principle provides a sharp, quantitative measure of how tightly probability is woven into the fabric of a quantum universe. To understand it, think of the prix-fixe menus in certain Chinese restaurants. Dishes are arranged in two columns, A and B, and if, for example, you order the first dish in column A, you are not allowed to order the first dish in column B; if you order the second dish in column A, you are not allowed to order the second dish in column B, and so forth. In this way, the restaurant has set up a dietary dualism, a culinary complementarity (one, in particular, that is designed to prevent you from piling up the most expensive dishes). On the prix-fixe menu you can have Peking Duck or Lobster Cantonese, but not both.

Heisenberg's uncertainty principle is similar. It says, roughly speaking, that the physical features of the microscopic realm (particle positions, velocities, energies, angular momenta, and so on) can be divided into two lists, A and B. And as Heisenberg discovered, knowledge of the first feature from list A fundamentally compromises your ability to have knowledge about the first feature from list B; knowledge of the second feature from list A fundamentally compromises your ability to have knowledge of the second feature from list B; and so on. Moreover, like being allowed a dish containing some Peking Duck and some Lobster Cantonese, but only in proportions that add up to the same total price, the more precise your knowledge of a feature from one list, the less precise your knowledge can possibly be about the corresponding feature from the second list. The fundamental inability to determine simultaneously all features from both lists—to determine with certainty all of these features of the microscopic realm—is the uncertainty revealed by Heisenberg's principle.